DT124A - Lab 2
From Karnaugh Maps and SoP, PoS to a Realised Circuit

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Authorship Statement

Pranish and Gagan declare that this lab report is a result of their own original work. Any outside sources used have been properly cited and acknowledged. This report has not been submitted previously for any academic or professional purpose.

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1 Introduction

This lab was a part of DT102A fall semester 2025 and was held on campus in Kristianstad. The objective of the lab was to show proficiency in going from a Karnaugh map through simplification using Boolean algebra and then getting a simplified expression that was realised on breadboard with correspondingcircuit. The assignment was divided in two different parts. The first task is a real- isation from Karnaugh maps using either NAND or NOR gates showcasing that both type of logic gates, and therefore logical operators, can express the same logic. The second task is a realisation of a two bit adder showing simplification of longer logical expressions using Karnaugh maps for simpli- fication.

2 Materials and Methods

The instruction was a published pdf-file where step by step instruction by creating a Karnaugh map, simplification and then this was implemented. This was available on the student platform Canvas before the lab exercise. The material used was the assigned lab training breadboard system, several 4011 and 4001 ICs and wire for connection.

3 Result

3.1 Karnaugh map

A Karnaugh map (K-map) is a visual method used in digital logic design to simplify Boolean algebra expressions. It helps minimize logic functions without having to use Boolean algebra rules directly, making it easier to design digital circuits efficiently.

3.2 What is this report about?

This report includes simplification of Boolean functions using K-maps, SOP and POS forms, and practical implementation with NAND, NOR, AND, OR, and NOT gates. It also covers the design of a 2-bit multiplier circuit.

3.3 Lab Question 1: Karnaugh Map Simplification - SOP Form

Table 1: Truth Table for the function to be implemented in Task 1

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

This truth table represents $F(A, B, C)$ which is to be represented in both SOP and POS forms later.

3.3.1 Objective

Given a Boolean function defined by its truth table, fill in the 1s in the Karnaugh map, group adjacent 1s, and derive the minimal SOP (Sum of Products) expression.

3.3.2 Karnaugh Map

C \ AB	00	01	11	10
0	0	1	0	0
1	1	1	0	1

Figure 1: 3-variable Karnaugh Map for $F(A, B, C)$

This is the Karnaugh map of the function represented by Table 1. We get the below minimized SOP expression from this K-map*(Figure 1).*

3.3.3 Minimized SOP expression

From the K-map groupings, the minimal Sum of Products (SOP) expression is:

$$F(A, B, C) = \overline{B}C + AB$$

3.4 Lab Question 2: SOP implementation using only NAND gates

3.4.1 Objective

Draw a circuit that shows how the expression can be solved using only 4 NAND gates with two inputs each, and verify the truth table.

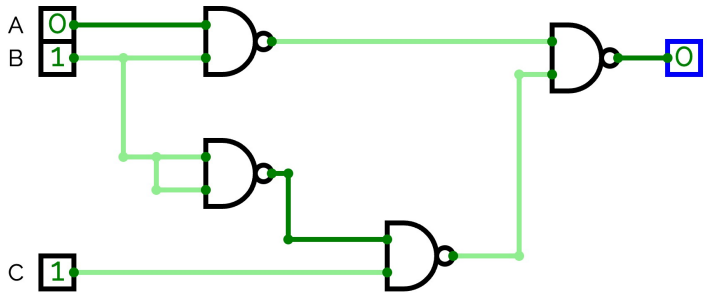


Figure 2: NAND-only implementation of $F(A, B, C)$. Source: Made by the authors.

Figure 2 is the NAND-only implementation of the SOP form of $F(A, B, C)$ in *circuitverse.org*. On comparison, the truth table from the lab matched the given function.

3.5 Lab Question 3: NAND-only circuit with IC pinout

Using the minimized SOP $X = AB + \overline{B}C$, we can implement the circuit with four NAND gates (IC 4011). From the pin diagram of IC 4011 (Figure 3), we can draw the following conclusion: one gate is used as inverter for B, two gates form $\overline{AB}$ and $\overline{\overline{B}C}$, and the last gate acts as the OR-combination. Pin connections follow the CD4011 datasheet.

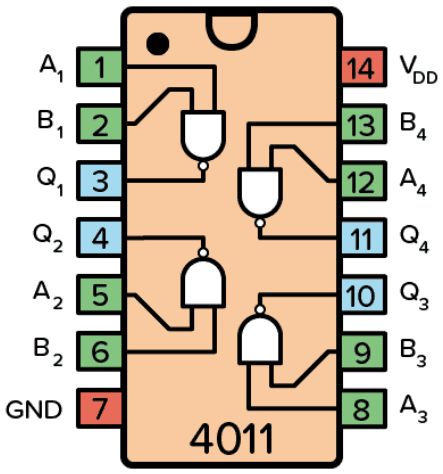


Figure 3: Pinout of CD4011 (quad 2-input NAND). Source: build-electronic-circuits.com

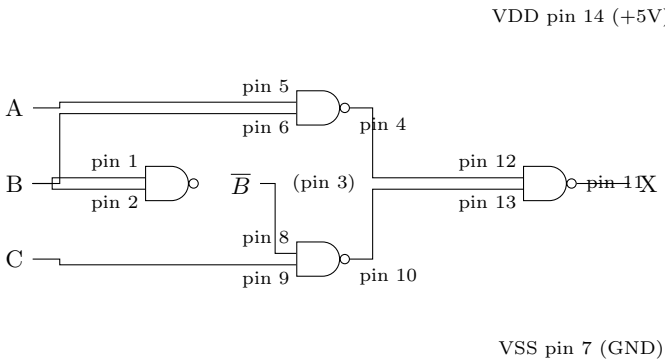


Figure 4: Implementation of $X = AB + \overline{B}C$ using CD4011 (quad NAND). Pin numbers shown. Source: Made in LaTeX

Table 2: Truth table for NAND-only implementation ($X = AB + \overline{B}C$)

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

3.6 Lab Question 4: Minimal POS form

3.6.1 Karnaugh Map

C \ AB	00	01	11	10
0	0	0		0
1		0		

Figure 5: 3-variable Karnaugh Map for $F(A, B, C)$

By grouping the 0s in the K-map we obtain the minimal POS expression:

$$(B + C)(A + \overline{B})$$

3.7 Lab Question 5 and 6: NOR-only implementation

The POS expression is implemented using four NOR gates (IC 4001). One gate inverts B, two compute $\overline{B + C}$ and $\overline{A + \overline{B}}$, and the last gate NORs them to yield X.

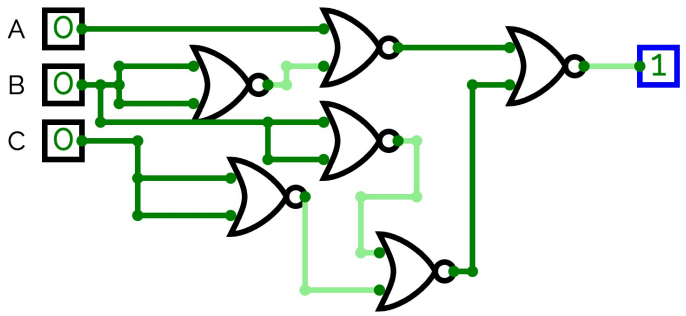


Figure 6: NOR-only implementation of $F(A, B, C)$. Source: Lab breadboard photo.

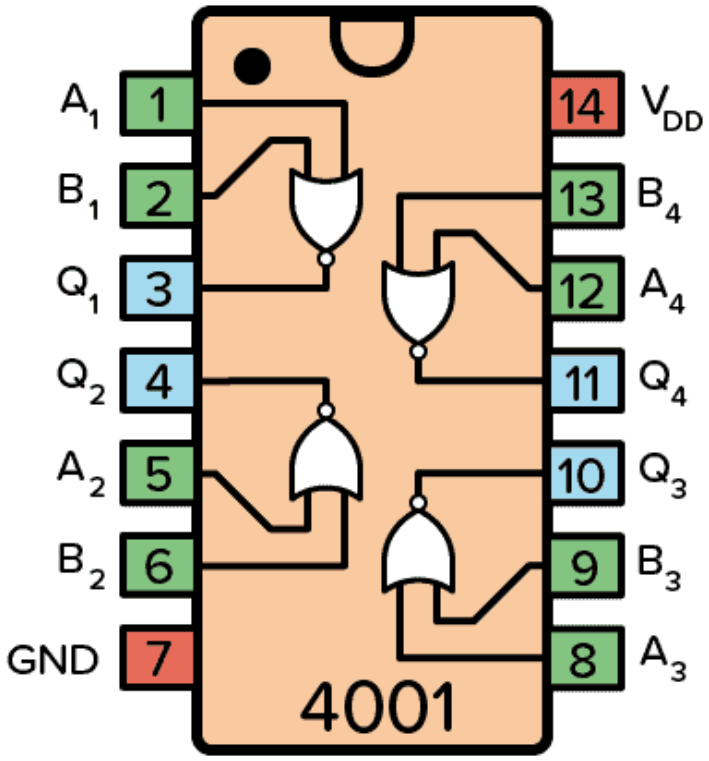


Figure 7: Pinout of CD4001 (quad 2-input NOR). Source: build-electronic-circuits.com

3.8 Lab Question 7: 2-bit multiplication truth table (summary)

Inputs: $K = (A, B)$ and $M = (C, D)$. Outputs: $N = (X, Y, Z, W)$. Only non-zero product rows are shown here:

	Inputs				Outputs				
	A	B	C	D	W	X	Y	Z	
0×0	0	0	0	0	0	0	0	0	0
0×1	0	0	0	1	0	0	0	0	0
0×2	0	0	1	0	0	0	0	0	0
0×3	0	0	1	1	0	0	0	0	0
1×0	0	1	0	0	0	0	0	0	0
1×1	0	1	0	1	0	0	0	1	1
1×2	0	1	1	0	0	0	1	0	2
1×3	0	1	1	1	0	0	1	1	3
2×0	1	0	0	0	0	0	0	0	0
2×1	1	0	0	1	0	0	1	0	2
2×2	1	0	1	0	0	1	0	0	4
2×3	1	0	1	1	0	1	1	0	6
3×0	1	1	0	0	0	0	0	0	0
3×1	1	1	0	1	0	0	1	1	3
3×2	1	1	1	0	0	1	1	0	6
3x3	1	1	1	1	1	0	0	1	9

Figure 8: Truth Table for the function to be implemented in Task 2

3.9 Lab Question 8: K-map outline for 2-bit multiplier

3.9.1 Karnaugh Maps

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	1	0
10	0	0	0	0

Figure 9: 4-variable Karnaugh Map for X

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	1
10	0	0	1	1

Figure 10: 4-variable Karnaugh Map for Y

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	1	1
11	0	1	0	1
10	0	1	1	0

Figure 11: 4-variable Karnaugh Map for Z

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	1	1	0
11	0	1	1	0
10	0	0	0	0

Figure 12: 4-variable Karnaugh Map for W

3.10 Lab Question 9: Minimized SOP expressions

$$X = ABCD$$

$$Y = AC(\overline{B} + \overline{D})$$

$$Z = AD\overline{B} + AD\overline{C} + BC\overline{A} + BC\overline{D}$$

$$W = BD$$

3.11 Lab Question 10: Circuit diagram implementation

ICs required

- 7404 (hex inverter)
- 7408 (quad 2-input AND)
- 7432 (quad 2-input OR)

Implementation notes

- Use 7404 to generate complements ($\overline{A}, \overline{B}, \overline{C}, \overline{D}$).
- Use 7408 ANDs to implement product terms. Cascade ANDs for 3- and 4-input terms.
- Use 7432 ORs to sum product terms. Cascade when more than 2 terms.

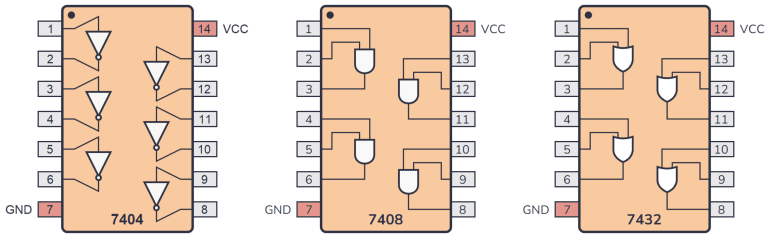


Figure 13: Pinouts of 7404, 7408, 7432 ICs. Sources: build-electronic-circuits.com

4 Lab Question 11: Discussion and Reflection

4.1 Hardest Part

The most challenging aspect of the circuit was designing the Z output. This output required four distinct product terms, several of which included inverted variables. As a result, the wiring became more complex and difficult to organize, increasing the chance of mistakes during implementation.

4.2 Did It Work Immediately?

The circuit did not function correctly on the first attempt. Several output errors were observed, which were later traced back to swapped connections and incorrect pin usage on the logic ICs.

4.3 Troubleshooting Steps

1. The student first verified the outputs of each inverter individually to ensure proper signal inversion.
2. Next, each product term generated by the AND gates was tested using LEDs to confirm their correctness.
3. The OR gate connections were then reviewed systematically to verify that the product terms were combined as intended.
4. Finally, any incorrect wiring was corrected, and the circuit was re-tested against the expected truth table to confirm proper functionality.

5 Bibliography

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